A Dynamic Weighted Fusion Model for Multimodal Sentiment Analysis

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Abstract

Multimodal sentiment analysis (MSA) tasks leverage diverse data sources, such as textual, acoustic, and visual data, to infer users' sentiment states. Previous research has mainly focused on capturing the differences and consistency of sentiment information between different modalities, emphasizing cross-modal interaction, while neglecting the in-depth exploration of sentiment information within individual modalities. Additionally, existing MSA methods rarely examine the contribution of each modality to model performance. To address these issues, our paper proposes a dynamic weighted fusion model for multimodal sentiment analysis. Specifically, we first design a Multi-Level Semantic Enhancement module (MLSE) for each single modality, aiming to extract semantic information from multiple perspectives and levels within the single mode. Subsequently, we utilize genetic algorithms to dynamically calculate the optimal weight parameters for each modality, thereby evaluating the impact of each modality on model performance. We conduct extensive experiments on three benchmark datasets (CMU-MOSI, CMU-MOSEI and CH-SIMS), and the results demonstrate that our proposed model outperforms state-of-the-art models across various metrics.

Keywords

Multimodal sentiment analysis; Multimodal interaction; Machine learning; Multimodal fusion;

1. Introduction

With the proliferation of information technology and social media, user-generated data displays increasingly diverse characteristics, providing a valuable resource for multimodal sentiment analysis [1], [2]. Multimodal sentiment analysis (MSA) integrates multiple sensory modalities, including textual, acoustic, and visual elements, to infer users' emotions and attitudes [3]. Traditional text-based sentiment analysis primarily relies on identifying emotional vocabulary and phrases in text to infer user emotions [4], [5]. However, human emotions are often complex, involving speech intonation, facial expressions, gestures, and other nonverbal cues. Traditional text-based sentiment analysis struggles to address the intricacies of this data, and sentiment analysis research is gradually shifting from unimodal to multimodal approaches [6]. By incorporating these diverse emotional expressions, multimodal sentiment analysis can more accurately and comprehensively capture user emotions, thereby improving the reliability and effectiveness of sentiment prediction.

Previous works have made significant progress in exploring the complementarity between multimodal data [7], [8], [9]. However, prior research on multimodal sentiment analysis (MSA) has predominantly focused on capturing the differences and consistency of emotional information across various modalities, emphasizing the interaction between these modalities [9], [10]. This has often come at the expense of a thorough exploration of emotional information within a single mode. Additionally, current MSA methods seldom address the specific contributions of individual modes to the overall model performance. When information from different modes is inconsistent or mismatched, accurately and effectively identifying user emotions poses a significant challenge [10]. For instance, as shown in Fig 1, the text is expressed as *"And um while the tracker made the film look real nice and cute"*. From words *"real nice"* and *"cute"*, we can extract positive and positive emotions, but the facial expression can be inferred as neutral or even negative, and the corresponding emotion label is negative. In this case, relying on cross modal interaction alone is not enough to compensate for the loss of potential information or misleading clues introduced by a single mode, and the inconsistency of modal information poses a complex challenge to emotion recognition. Therefore, effectively integrating information from various modes, exploring the weight ratio between each mode, and more accurately identifying the emotional state of users have become urgent issues that need to be addressed.

To address the issue of multimodal sentiment representation mentioned above, we propose a dynamic weighted fusion framework for multimodal sentiment analysis. Specifically, we design a Multi-Level Semantic Enhancement module (MLSE), which leverages convolutional neural networks [11] and multi-head attention mechanisms [12] to capture abstract information and contextual relationships within each modality, thereby enhancing the representational capacity of each modality at multiple levels. Then, we use genetic algorithms to optimize the weight parameters of each modality, maximizing the contribution of each modality's emotional features to the overall model performance. Through the iterative optimization process of the genetic algorithm, the model can automatically adjust the modal weights to flexibly adapt to changes in modal information across different environments, calculating the most suitable weights for each modality. Lastly, we utilize weighted fusion strategy to derive the final emotional prediction outcomes. This approach enhances the semantic information of each modality at multiple levels by deeply mining individual modalities. Additionally, when modal information is inconsistent or mismatched, the model can calculate the similarity between each modality and the true sentiment labels, dynamically adjusting each modality's weight ratio and contribution. This provides innovative solutions for the field of multimodal sentiment analysis. Our primary contributions can be outlined as follows:

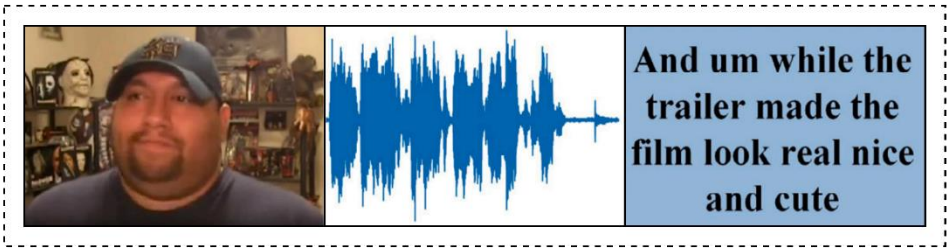

Fig 1. Inconsistent emotions expressed between modalities

1. We have designed a multi-level semantic enhancement module that uses convolutional neural networks and self-attention mechanisms to capture abstract information and contextual relationships across modalities, enhancing the representational capabilities of each modality at multiple levels.

2. We introduce a genetic algorithm suitable for multimodal sentiment analysis, which dynamically evolves the weight parameters of each modality in different contexts to improve the accuracy and stability of sentiment analysis.

3. Extensive experiments conducted on three benchmark datasets demonstrate that our proposed method exhibits superior performance compared to existing advanced models.

2. Related work

2.1 Multimodal Sentiment Analysis

In recent years, the field of multimodal emotion analysis has attracted more and more attention [13]. Multimodal emotion analysis combines information from different sensory modes, such as text, voice, image, etc., to infer and understand users' emotions, attitudes and emotional tendencies [14], [15]. In comparison to traditional sentiment analysis methods based on a single mode, multimodal sentiment analysis can capture and comprehend users' emotions more comprehensively **错误!未找到引用源。**. In their early work, Zadeh et al. [7] proposed to integrate all modal information into high-dimensional tensors by using triple Cartesian product operation and LSTM network [17], but this method has high computational complexity. In order to simplify this complex calculation, Liu et al. [8] proposed a low rank tensor fusion network (LMF) to improve the efficiency of data processing by reducing the parameters of the low rank tensor. In order to strengthen the integration of cross modes, Wang et al. [18] proposed a hierarchical fusion network (Raven) with both local fusion and global fusion. This network captures the fine-grained relationship between modes through the sliding window mechanism while maintaining computational efficiency. With the proposal of transformer [12], more and more researchers apply it to MSA tasks. Tsai et al. [19] utilized directional pairwise cross-modal attention mechanisms to process information interaction between different modalities, and proposed Multimodal Transformer (MulT). MulT can effectively handle unaligned multimodal language sequences and learn the correlations and interactions between different modalities. The TEDT model proposed by Wang et al. [20] introduces a multimodal encoding decoding network, which takes the text mode as the main information to improve the performance of the model. In recent research, Zhang et al. [21] proposed the ICDN model to solve the time series alignment and missing modal information by introducing a cross-modal converter. Wang et al. [22] proposed a text enhancement converter fusion network (TETFN), which focuses on learning text oriented pairwise cross modal mapping to obtain a unified and robust multimodal representation. Yao et al. [23] proposed a multi-odal feature decoupling strategy that categorizes sentiment features into common and private features (FDR-MSA), the important information in the decoupling process is retained and the influence of redundant data is reduced by introducing a dual feature reconstruction mechanism composed of unimodal feature reconstruction (UFR) and multi-modal feature reconstruction (MFR).

2.2 Modalities fusion strategy

Effectively integrating features from different modalities is critical in multimodal sentiment analysis [24]. The current multimodal fusion strategies are primarily categorized into two groups: traditional methods and deep learning-based methods. Illustrated in Fig 2, traditional methods comprise early fusion and late fusion, while deep learning-based methods are further classified into attention-based fusion and attention-unrelated fusion.

In early fusion strategies, features from different modalities are merged at the input layer, streamlining the feature fusion process. Self-MM [25] designed a label generation module based on a self-supervised learning strategy, dividing multimodal sentiment analysis tasks into multimodal tasks and single modal subtasks. Joint training is implemented on both multimodal and single modal tasks to learn the consistencies and discrepancies between modalities. Late fusion, commonly referred to as decision-level fusion, entails the process of learning by independently introducing features from various modalities into their designated models. Subsequently, these modal-specific features are integrated at a more advanced stage within the model. The primary methods encompassed in this approach are averaging [26], majority voting [27], weighted sum [28], etc. Late fusion tends to lack consideration for the interaction information between different modalities, which may lead to overlooking the complementarity and correlation between these modalities. Attention-based fusion strategies typically utilize various attention mechanisms to achieve information exchange between or within modalities. MARN [29] utilized the Multi-attention Block (MAB) to discover interactions between modalities and store them within the hybrid memory of a recurrent component called the Long-Short Term Hybrid Memory (LSTHM). Ou et al. [30] proposed a multimodal local global attention network (MMLGAN) for emotional video content analysis extends the attention mechanism to multi-level fusion and designs a multimodal fusion unit to obtain a global representation of emotional videos. Attention-based methods focus more on the correlation and consistency between different modalities, which may lead to the model ignoring the differences between modalities when weighting different modal features.

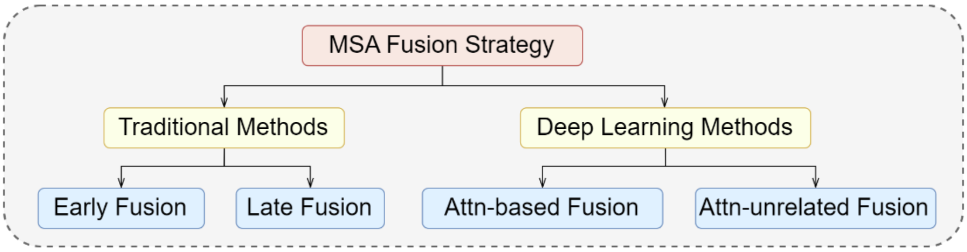

Fig 2. MSA modalities fusion strategy

3. Methodology

This section first introduces the task definition, and then details the proposed model framework. Figure 2 shows the overall structure of the proposed model. We will elaborate on the proposed model from four aspects: feature extraction, feature enhancement, weight calculation, and weighted fusion.

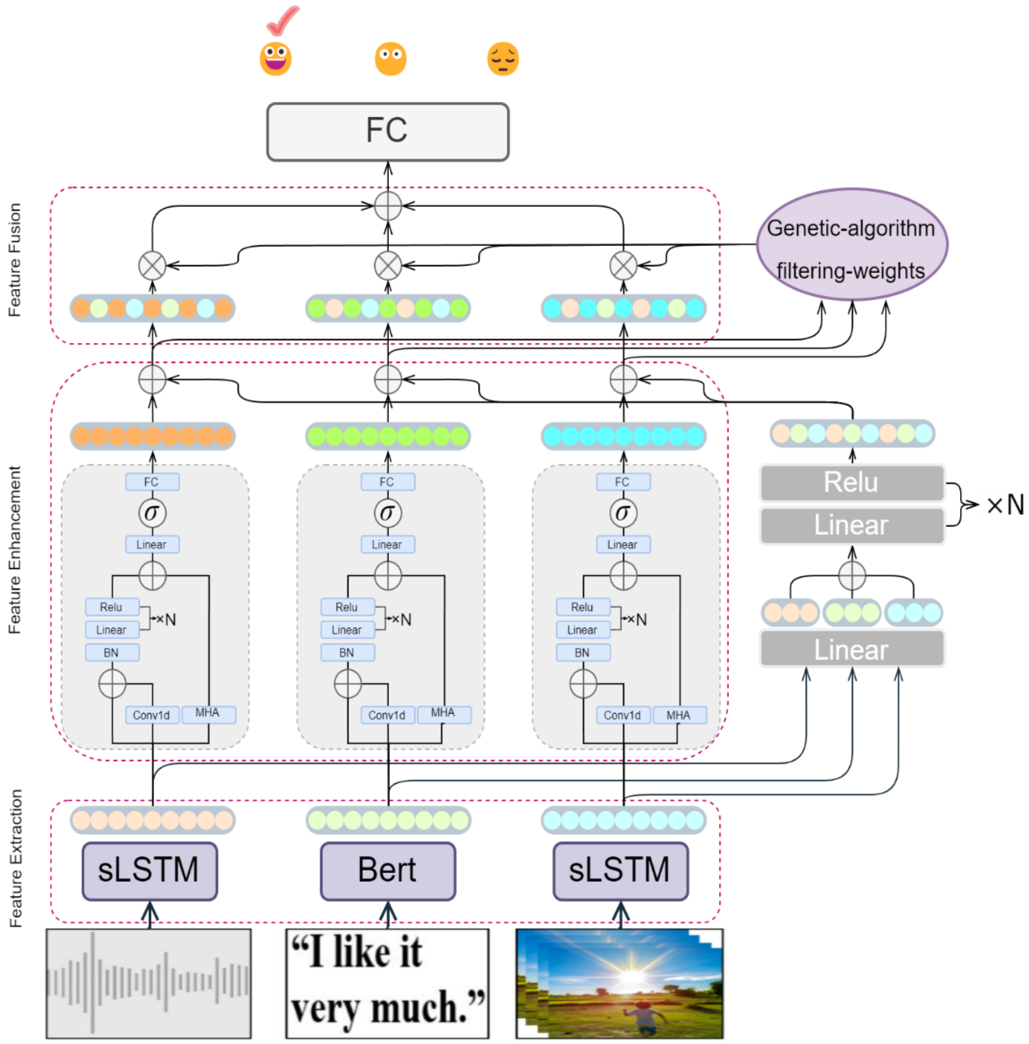

Fig 3. Overall architecture of the proposed Model

3.1Task definition

The task of Multimodal Sentiment Analysis (MSA) aims to deduce and identify the emotional polarity conveyed by users through the analysis of various modalities of information. The benchmark dataset used in this study divides video segments into text (t), audio (a), and visual (v) signals, and these feature signals of different modalities will serve as inputs to the model. The symbol is represented as $U_m \in \mathbb{R}^{l_m \times d_m}$, where l_m denotes the sequence length, d_m denotes the feature dimension, $m \in \{t, a, v\}$. Correspondingly, the model outputs an emotion intensity result Y^p. Specifically, $Y^p > 0$ indicates positive sentiments, $Y^p < 0$ indicates negative sentiments, and $Y^p = 0$ indicates neutral sentiments.

3.2 Feature preprocessing and extraction

Following previous research [9], [25], the pre-trained BERT [31] is proficient in capturing bidirectional contextual information and rich semantic relationships between different vocabulary, generating sentence representations imbued with rich sentiment information. Therefore, we select the BERT model which consisting of 12 stacked transformers as the text encoder, and the first word vector of the last layer of BERT to represent the output features. The representation of the resulting text modality is

$$F_t = BERT\left(U_t;\ \theta_t^{bert}\right) \in \mathbb{R}^{d_t}$$

where U_t represents the original text data, θ_{bert} denotes all learnable parameters of the pre-trained BERT, and d_t is the feature dimensions.

For visual modality, the three datasets use OpenFace2.0 toolkit [32] for initial video processing, including extracting facial action units, facial markers, head posture, line of sight tracking, and HOG functions [38], [40]. Finally, 709-dimensional frame-level visual features are extracted by CH-SIMS [33], 35-dimensional by CMU-MOSI [34] and CMU-MOSEI [35]. Then we use LSTM network [36] to process the preprocessed visual features U_v which provided by these three datasets. Long-Short-Term-Memory (LSTM) is a commonly used variant of Recurrent Neural Network (RNN) [37], which has stronger memory ability, solves the problem of gradient vanishing, and is more suitable for processing long sequence data. We use the hidden state of the final time step of LSTM to represent the visual modality, which contains information about the entire input sequence and can be used for the final prediction or further processing of the network.

$$F_v = LSTM(U_v;\ \theta_v^{lstm}) \in \mathbb{R}^{d_v}$$

For audio modality, the datasets CMU-MOSI and CMU-MOSEI use the speech processing tool COVAREP [39] to extract feature parameters, such as Mel-frequency cepstral coefficients (MFCCs), volume, glottal source, and other features related to emotion and intonation. These feature parameters can be obtained through the CMU Multimodal SDK. The dataset CH-SIMS uses the LibROSA speech toolkit [41] to extract 33-dimensional acoustic features at 22050Hz with default parameters, which are related to emotions and mood [42]. Consistent with visual feature processing techniques, we have also incorporated LSTM to extract and analyze the preprocessed acoustic features,

$$F_a = LSTM(U_a;\ \theta_a^{lstm}) \in \mathbb{R}^{d_a}$$

where U_v and U_a denote the preprocessed audio and visual features respectively, and θ_m^{lstm} represents all learnable parameters in LSTM, $m \in \{v, a\}$.

3.3Multi-Level Semantic Enhancement Fusion module

This section discusses the multi-level semantic enhancement module of the modality proposed in this study. After feature extraction in the previous section, we use the

Multi-Level Semantic Enhancement Fusion (MLSE) module to further enhance the semantic information of modal features from multi-levels. Specifically, we replicate the feature F_m obtained in the previous step into three duplicates. The first duplicate retains the original form, and the second undergoes processing through a one-dimensional convolutional network [11] to amplify the local representation of the modality. Conv1D layers are valuable in extracting important local patterns in sequential data, providing flexibility and efficiency in effectively modeling one-dimensional data structures. Next, we concatenate them and pass them through a Batch Norm layer, which helps improve the model's performance, accelerates the training process, and enhances the generalization ability of the model. Then, after passing through N linear layers, the output is obtained. The specific calculation process is shown in formulas:

$$F_m' = Conv1D(F_m, k_m)$$

$$F_m'' = BN(F_m \oplus F_m')$$

$$F_m^{local} = N \times RELU(Linear(F_m''); \theta_m)$$

where k_m denotes the kernel size for each modality, $BN(\cdot)$ denotes BatchNorm layer, θ_m denotes learnable parameters, $m \in \{t, v, a\}$.

We further utilize the multi-head attention to enhance the global semantic information of the third duplicate. Attention is proficient in capturing the dependency relationship between any two positions (tokens) in the input sequence, regardless of the distance between the two positions, which helps the model better understand the global structure and relationships in the input sequence,

$$F_m^{global} = MultiHead(F_m, F_m, F_m) = Concat(head_1, head_2, \cdots, head_k)W^o$$

$$head_i = Attention\left(F_m W_i^q, F_m W_i^k, F_m W_i^v\right)$$

where W^o, W_i^q, W_i^k and W_i^v are learnable weight matrices.

Finally, we use a gated network to fuse multi-level semantic information:

$$F_m^{output} = FC\left(\sigma\left(F_m^{local} \oplus F_m^{global}\right)\right)$$

where $\sigma(\cdot)$ and FC denote sigmoid activation functions and fully-connected layers respectively.

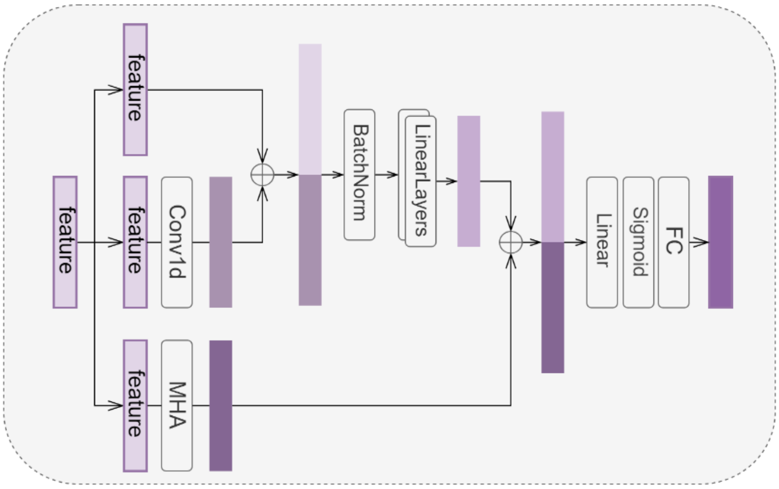

Fig 4. The Multi-Level Semantic Fusion Module

3.4 Dynamic weight evolution

Genetic algorithms (GA) are optimization algorithms inspired by biological evolution, used to search and optimize the solution space of problems.

Algorithm 1 The general process of genetic algorithm

Initialize population
while termination condition not met **do**
    Evaluate individuals
    Select parents
    Recombine selected parents
    Mutate offspring
    Replace population with new generation
end while
return Best individual

Algorithm 2 The Evaluate function of genetic algorithm

function Evaluate fitness (weights, data)
    text data, audio data, visual data, labels ← data
    text score ← weights [0] × text data
    audio score ← weights [1] × audio data
    visual score ← weights [2] × visual data
    layer ← a linear layer without bias
    text score, audio score, visual score ← apply layer
    labels ← extract numpy labels
    return sum of mean absolute errors between scores and labels
end function

3.5 Feature Fusion

3.6

4. Experiments

In this section, we empirically evaluated the performance of three commonly used public dataset models in multimodal sentiment analysis tasks. Specifically, this section focuses on the dataset used, evaluation metrics, baseline methods, and experimental details.

4.1 Datasets

This study used three common benchmark datasets, CMU-MOSI [34], CMU-MOSEI [35], and CH-SIMS [33], which are widely used in MSA tasks, to train and test our proposed model. The first two datasets are in English and the last one is in Chinese. The number of emotion categories for each dataset is shown in Table 1, and we have briefly introduced these datasets.

Table 1 Dataset statistics in MOSI, MOSEI, and SIMS.

Dataset	Train	Valid	Test	All
CMU-MOSI	1284	229	686	2199
CMU-MOEI	16326	1871	4659	22856
CH-SIMS	1368	456	457	2281

CMU-MOSI: The CMU-MOSI dataset is a multimodal dataset used for sentiment analysis and sentiment intensity prediction. This dataset mainly includes 93 short monologue video clips extracted from 2199 YouTube movie review videos. It covers facial expressions, speech, text, and contextual information. Each video clip is labeled with emotional labels, ranging from -3 (highly negative) to 3 (highly positive), used to indicate emotional tendencies and intensity.

CMU-MOSEI: This is an enhanced version of CMU-MOSI, significantly expanding its scale and scope. The dataset contains 22856 annotated video clips covering 250 different topics. This dataset has the same tag values as the CMU-MOSI dataset.

CH-SIMS: The CH-SIMS dataset is a Chinese MSA dataset that collects a total of 60 original videos from movies, TV shows, and variety shows. These videos are split into 2281 video clips, providing comprehensive multimodal annotations and individual annotations for each modality. Each sample in the dataset was assigned an emotional score ranging from -1 (indicating strong negativity) to 1 (indicating strong positivity).

4.2 Baselines

To comprehensively validate the performance and effectiveness of the proposed model, we compared our model with several baselines, including some classic models and the most recent state-of-the-art models. The models we compared are as follows:

MulT [19] (2019): Multimodal Transformer (MulT) utilizes directed pairwise cross modal attention to capture correlations between different modalities, enhance interactions between different modalities, and obtain richer multimodal feature representations.

ICCN [43] (2019): The Interaction Canonical Correlation Network (ICCN) learns correlations between all three modes via deep canonical correlation analysis (DCCA) to achieve deep correlations between text and audio, as well as text and video.

MAG-BERT [44] (2020): This model proposes to combine the core architecture of Bert and XLNet [45] with multimodal adaptation gates, and use non-linguistic modal information to adjust language modalities.

MISA [9] (2020): The Modal Invariant and Modal Specific Representation (MISA) divides each modality into modal invariant and modal specific subspaces and learns the modal representations of each subspace, providing a comprehensive perspective for multimodal data.

Self-MM [25] (2021): The model Utilizes self-supervised multi task learning to obtain single modal labels, achieving joint training of multimodal and single modal tasks, helps to learn consistency and differences between modalities.

MMIM [46] (2021): Multimodal Information Modulation (MMIM) is the first to introduce information maximization into multimodal sentiment analysis tasks, enhancing mutual information between input single modal pairs and between multimodal fusion results and single modal inputs through a hierarchical mutual information maximization framework.

HYCON [47] (2023): This method proposes a novel multimodal representation learning framework based on contrastive learning, which designs three types of losses to comprehensively learn inter-modal and intra-modal dynamics in both supervised and unsupervised ways.

SIMR [48] (2023): The speaker independent multimodal representation (SIMR) framework decomposes nonverbal representations into personal style encoding and emotional representation to mitigate the influence of personal style, and uses enhanced cross modal converters to explore contextual interactions between modalities.

MUTA-Net 错误!未找到引用源。 (2023): This modal discourse temporal attention network applies discourse level representation to interactions between different modalities, minimizing intra class distance and maximizing inter class distance, enhancing the discriminative power of representation.

TETFN [22] (2023): The Text Enhanced Transformer Fusion Network (TETFN) obtains effective unified multimodal representation by learning paired cross modal mapping based on text, and pays attention to modal differences through single modal labels.

AOBERT (Kim& Park, 2023) All modalities in One BERT network integrates three types of modal representations using a single stream Transformer and trains MMLM and AP tasks using multimodal representations.

ALMT [43] (2023): Adaptive Language Guided Multimodal Converter (ALMT) introduces an Adaptive Hyper modal Learning (AHL) module to learn irrelevant and conflicting information contained in visual and auditory features under the guidance of language features, in order to obtain complementary and joint multimodal representations.

DMD (2023): Decoupling Multimodal Distillation (DMD) preprocessing and extracting features of different modal information, utilizing knowledge distillation techniques to transfer and fuse knowledge between different modalities

MCR [47] (2023): The MCR method draws inspiration from human perceptual paradigms and proposes a collaborative reinforcement method for target and source patterns. This method aims to promote effective cross modal interaction and fusion across various levels of detail.

TMBL [41] (2024): A Transformer based multimodal binding and learning model (TMBL) combines bimodal and tri modal binding mechanisms, fine-grained convolution modules, and utilizes similarity and dissimilarity losses to promote model convergence.

4.3 Experimental parameters

We conducted extensive experiments to investigate the effects of hyperparameters and learning strategies to determine the optimal settings. We adopt the Adam optimizer (Kingma &Ba, 2015) with an initial learning rate of $5^e - 5$ to optimize trainable parameters. Considering computing resources and training efficiency, we have set the batch size to 32. Especially, the early stopping strategy is adopted during the training process, which indicates if the validation loss does not decrease for 8 consecutive epochs, the training process will stop. The model proposed in this article is based on Python 3.9 and Python 1.4.0 implementation. All experiments were conducted using a single GeForce RTX 4090 (24G) GPU under the Windows 10 operating system. The pre trained BERT model can be obtained from the Transformer toolkit published by Hugging Face. The hyperparameter settings during the training process are shown in Table 2.

4.4 Evaluation metrics

This study uses five different metrics to measure the experimental performance of the proposed model: MAE (mean absolute error) directly calculates the error between the true label and the model's predicted label; Corr (Pearson correlation coefficient) measures the degree of prediction bias. Acc7 (seven classification accuracies: sentiment labels between [-3,3]) is used for sentiment rating classification tasks; Acc2 (binary classification accuracy: positive/negative sentiment) and F1 score are used for binary sentiment classification tasks. Except for MAE, higher metric values indicate better model performance. The calculation formulas for these metrics are as follows:

5. Results and analysis

5.1 Quantitative analysis

Table 2 Performance comparison with previous models on CMU-MOSEI. "-": not presented in the original paper. All results are reported in the original papers. And the results with * are reproduced under the same conditions. The top three results are highlighted in boldface, and the best results are additionally underlined. In Acc-2 and F1-Score, the left of the "/" is calculated as "negative/non-negative" and the right is calculated as "negative/positive".

Models	MAE (↓)	Acc-2 (↑)	F1-score(↑)	Acc-7 (↑)	Corr (↑)
MulT o	0.871	- / 83.0	- / 82.8	40.0	0.698
ICCN o	0.862	- / 83.07	- / 83.02	39.01	0.714
MAG-BERT*	0.727	82.37 / 84.43	82.50 / 84.61	43.62	0.781
MISA o	0.783	81.8 / 83.4	81.7 / 83.6	42.3	0.761
Self-MM o	0.713	84.0 / 85.98	84.42 / 85.95	-	0.798
MMIM*	0.755	83.67 / 85.37	83.62 / 85.37	45.34	0.773
HYCON o	0.713	- / 85.2	- / 85.1	46.6	0.790
SIMR o	0.706	84.2 / 86.1	84.0 / 86.1	48.2	0.798
MUTA-Net o	0.730	83.1 / 85.0	83.0 / 85.0	-	0.793
TETFN o (2023)	0.717	84.05/86.10	83.83/86.07	-	0.800
AOBERT o	0.856	85.2 / 85.6	85.4 / 86.4	40.2	0.700
ALMT*	0.720	82.94 / 84.91	83.07 / 84.97	47.08	0.783
DMD*	0.736	- / 85.4	- / 85.4	46.34	0.784
MCR o	0.824	- / 84.8	- / 84.6	42.6	0.735
TMBL o	0.867	81.78/83.84	82.41/84.29	36.3	0.762
Ours	0.695	84.84/86.69	84.83/86.72	48.52	0.810

Table 3 Performance comparison with previous models on CMU-MOSEI. "-": not presented in the

original paper. All results are reported in the original papers. And the results with * are reproduced under the same conditions. The top three results are highlighted in boldface, and the best results are additionally underlined. In Acc-2 and F1-Score, the left of the "/" is calculated as "negative/non-negative" and the right is calculated as "negative/positive".

Models	MAE (↓)	Acc-2 (↑)	F1-score(↑)	Acc-7 (↑)	Corr (↑)
MulT o	0.580	- / 82.5	- / 82.3	51.8	0.703
ICCN o	0.565	- / 84.18	- / 84.15	51.58	0.713
MAG-BERT*	0.543	82.5 / 84.82	82.77 / 84.71	52.67	0.755
MISA o	0.555	83.6 / 85.5	83.8 / 85.3	52.2	0.756
Self-MM o	0.530	82.81 / 85.17	82.53 / 85.30	-	0.765
MMIM*	0.543	83.08 / 85.04	83.48 / 84.88	45.34	0.758
HYCON o	0.601	- / 85.4	- / 85.6	52.8	0.776
SIMR o	0.580	82.5 / 82.9	81.9 / 82.9	52.6	0.696
MUTA-Net o	0.544	82.4 / 85.0	82.7 / 84.9	-	0.760
TETFN o (2023)	0.551	84.25 / 85.18	84.18 / 85.27	-	0.748
AOBERT o	0.515	84.9 / 86.2	85.0 / 85.9	54.5	0.763
ALMT*	0.533	82.23 / 85.66	81.89 / 85.79	52.82	0.777
DMD*	0.541	- / 85.5	- / 85.4	52.5	0.762
MCR o	0.554	- / 84.7	- / 84.3	54.5	0.736
TMBL o	0.545	84.23/85.84	84.87/85.92	52.4	0.766
Ours	0.526	83.69/85.97	83.86/85.98	54.8	0.768

5.2 Ablation study

Conclusion

This paper proposes a multimodal sentiment analysis model with a hybrid of three fusion methods for multimodal sentiment analysis tasks. This model combines the advantages of early fusion and model level fusion, and obtaining strong inter-modal connection features between modalities through multi type and multi-level fusion. Extensive experiments on CMU-MOSI and CMU-MOSEI datasets have shown that it has advantages over other methods in multimodal sentiment analysis tasks. Empirically, textual features are usually superior to non-textual features. In future work, we will further explore the contribution of strong and weak features in multimodal sentiment analysis tasks.

Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Authors contribution statement

Liang Yang: Conceptualization, Methodology, Software, Formal Analysis, Writing - Original Draft. **Xinhua Zhu**: Data Curation, Supervision, Funding Acquisition, Writing - Review & Editing. **Siyang Li**: Investigation, Resources, Project Administration.

Ethical and informed consent for data used

The datasets used in this study were CMU-MOSI and CMU-MOSEI created by researchers from Carnegie Mellon University. These datasets are publicly accessible and designed for academic use in the field of multimodal sentiment analysis. They do not contain any personally identifiable information or sensitive data. The use of these datasets complies with the terms and conditions set forth by their respective publishers. No specific informed consent was required for this study as the data used were obtained from publicly accessible sources which do not contain information related to individual human subjects. These data sources are open for research purposes and are anonymized in such a manner that individual privacy is maintained.

Data availability and access

The datasets used and analysed during the current study are available from the corresponding author.

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